

ASSIGNMENT 2, Question 1

1. Discussion on the Choice of Number of Folds (k)

For this experiment, a **10-fold Cross-Validation (k=10)** is the recommended choice. This decision is justified by the following factors:

- **Dataset Size:** The provided training dataset consists of **800 samples**. With $k=10$, each fold contains 80 samples for validation and 720 for training. This ensures the validation set is large enough to be statistically representative, while the training set (90% of the data) is large enough to closely approximate the performance of the model trained on the full dataset.
- **Bias-Variance Trade-off:**
 - **Low k (e.g., k=2):** Results in high bias because the model is trained on significantly less data (only 50% in a 2-fold split).
 - **High k (e.g., Leave-One-Out):** Results in high variance and high computational cost because the models are very correlated (trained on nearly identical data).
 - **k=10:** Offers a balanced compromise. It provides a low-bias estimate of the model's true error without the excessive computational cost of Leave-One-Out validation.

2. Comparison of "Err" Estimates: Validation vs. CV Approaches

When comparing the "Err" (typically summarized by the Residual Standard Error (RSE) and R^2) obtained from the Validation approach versus the Cross-Validation approach, you should highlight the following differences in reliability:

- **The Validation Approach (Single Split):**
 - **Volatility:** The error estimate in the validation approach is highly sensitive to how the data was randomly split. If the specific subset selected for the "test" partition happens to contain "easy" examples (points close to the regression line), the reported error will be optimistically low. Conversely, if it contains outliers, the error will be pessimistically high.
 - **Data Usage:** This method effectively "wastes" a portion of the available training data during the evaluation phase, as the model is evaluated on data it has never seen, but that data does not contribute to learning the parameters for that specific estimate.
- **The Cross-Validation (CV) Approach:**
 - **Robustness:** The CV approach provides a more reliable estimate of the model's generalization performance ("Err"). By averaging the results over k iterations, it smooths out the randomness associated with any single data split.
 - **Comprehensive Evaluation:** Unlike the single validation split, CV ensures that every single data point in the training set is used for both training and validation

exactly once. This reduces the risk that the performance metric is skewed by a specific, unrepresentative subset of the data.

Conclusion Report: it is observe that the CV estimate is different from the Validation estimate. And therefore concludes that the **CV estimate is the more trustworthy metric** for gauging how the model will perform on the unseen "independent test" dataset, as it minimizes the variance caused by random data splitting.

2. Python Code Solution

This script performs the grid search, generates the required plot, and prints the comparison metrics.

3. Python Code Solution (Question 3)

This script performs the Lasso experiment, generates the tuning plot, and prints the comparison with Ridge and OLS.

Note: I have set `max_iter=10000` for the Lasso model. Lasso optimization can sometimes be slow to converge on unscaled data, so increasing iterations helps avoid convergence warnings.

Also The experiment was repeated using a multivariate **Lasso (Least Absolute Shrinkage and Selection Operator) Regression** model. Similar to the Ridge experiment, the regularization parameter alpha was tuned using a **Grid Search** with **10-fold Cross-Validation** on the training dataset.